VPC-VPC Peering-VPC

Notes: Create a Jira ticket for this assignment, work on the tasks listed below, and submit the task completion document by EOD with clear steps and required evidence (outputs/screenshots)

Create one VPC, with 1 public subnet and 1 private subnet.

Enable VPC peering for cross-region.

Enable VPC peering for cross-account (you can collaborate with your friend to do this task).

Set up a VPC Transit Gateway.

Set up a VPC Endpoint.

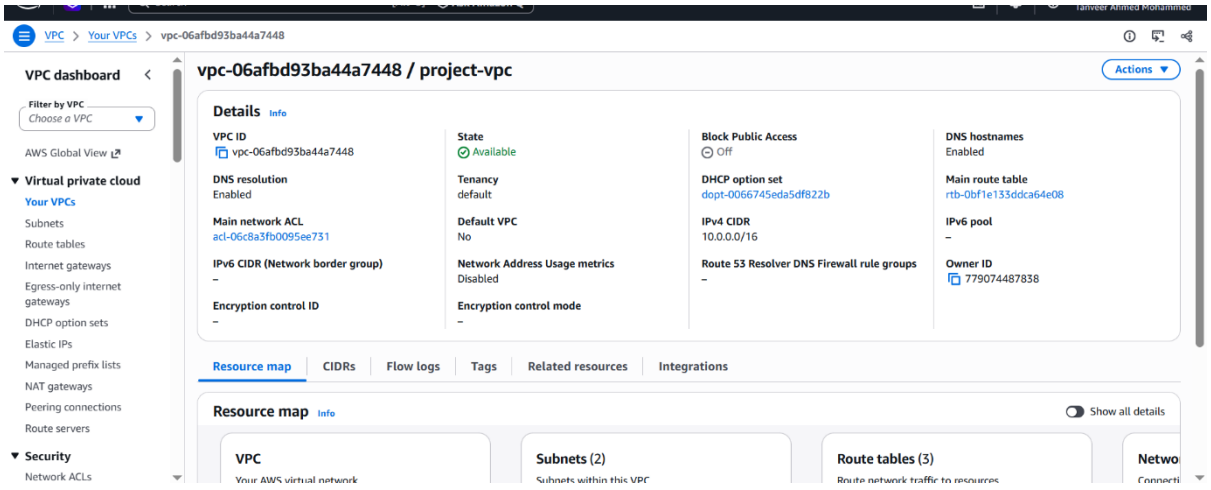
1) **Create one VPC, with 1 public subnet and 1 private subnet.**

Using the AWS Management Console

1.Create the VPC:5 min.

Open the Amazon VPC console, click **Create VPC**, and select **VPC and more** (this automates subnet and route table creation).

- Name your VPC (e.g., my-custom-vpc).
- Set the IPv4 CIDR block to 10.0.0.0/16.
- Set the Number of Availability Zones (AZs) to **1**.
- Set the Number of public subnets to **1** and private subnets to **1**.
- Set NAT gateways to **In 1 AZ** (this ensures your private subnet can reach the internet for updates).
- Click **Create VPC**.
  Verification: The dashboard displays a success message, and your VPC status shows as available.

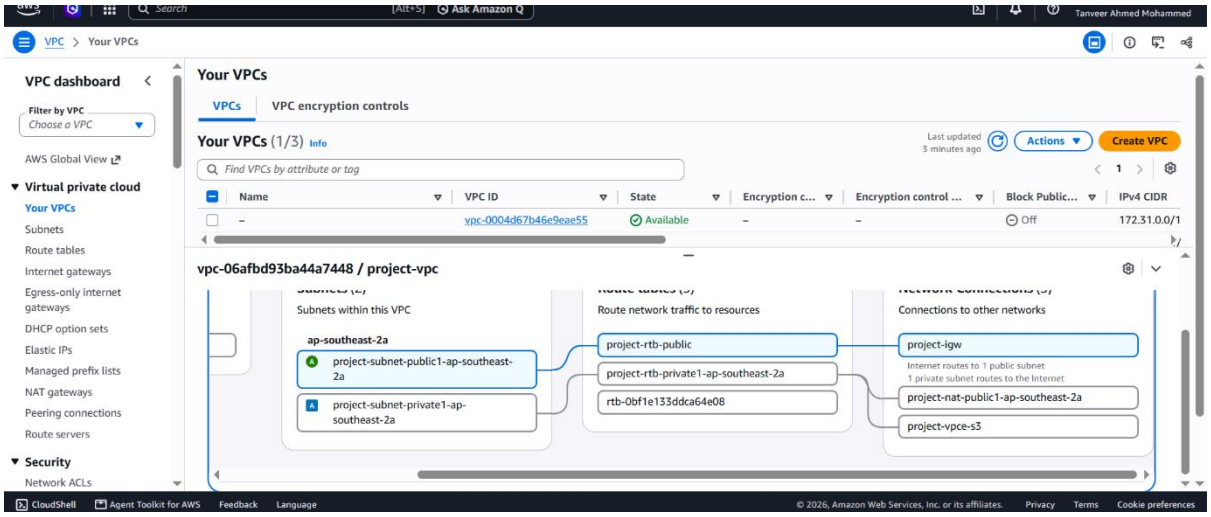


2.Verify Public Subnet Routing:2 min.

Navigate to Subnets, select your public subnet, and check the Route table tab.

- Confirm that the route table includes a route where destination 0.0.0.0/0 points to an Internet Gateway (igw-...).
- Ensure Auto-assign public IPv4 address is enabled on the public subnet so instances launched there receive a public IP automatically.

Verification: The subnet details page explicitly lists the Internet Gateway under routes.

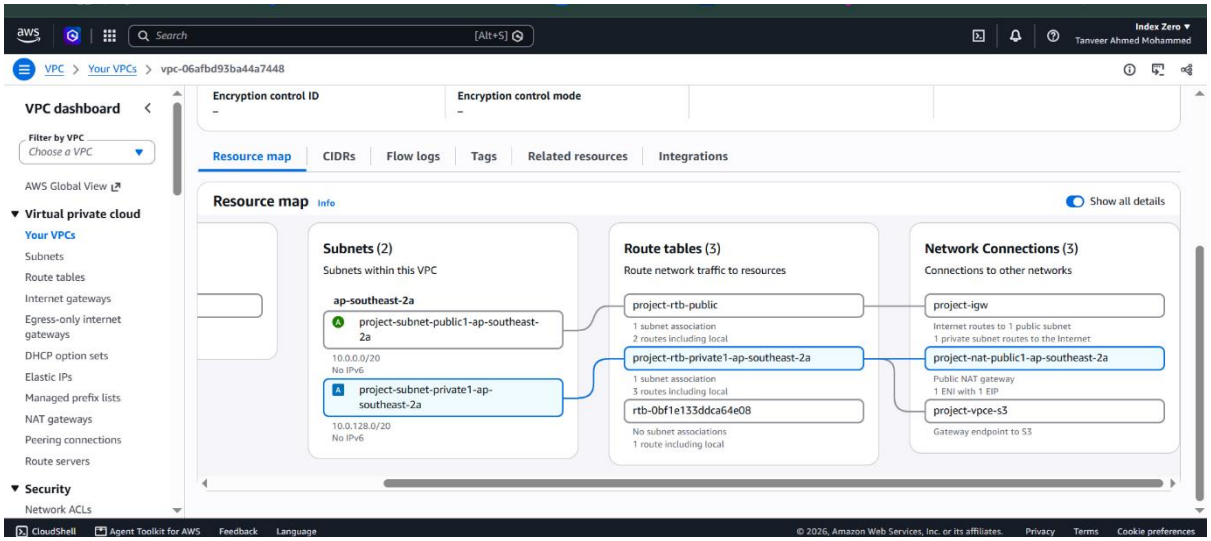


3.Verify Private Subnet Routing:2 min.

Select your private subnet and check its Route table tab.

- Confirm that the route table includes a route where destination 0.0.0.0/0 points to a NAT Gateway (nat-...).

Verification: The route table points outbound traffic from the private subnet to the NAT Gateway in the public subnet rather than directly to an Internet Gateway.



3.Check the Private Subnet Route Table:Private Subnet Routing Check.

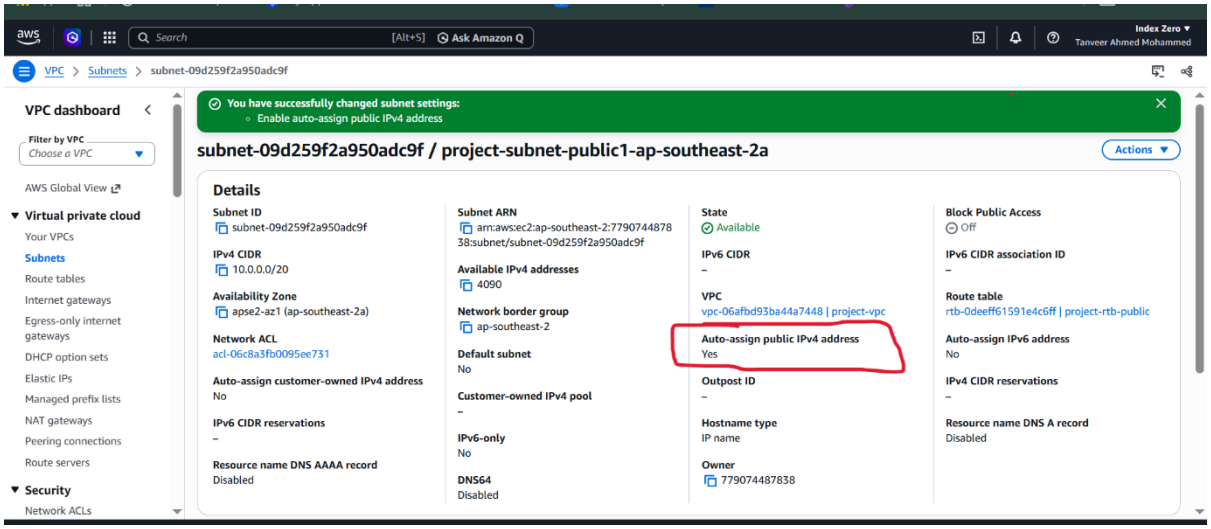
Select the route table associated with your private subnet, and click the Routes tab.

- *Verification:* You should see a local route (10.0.0.0/16 pointing to local) and a route where destination 0.0.0.0/0 points to your NAT Gateway (nat-...), not the Internet Gateway.

4.Confirm Public Subnet Auto-Assign IP:Subnet Settings Check.

Click Subnets, select your public subnet, and check the Details tab.

- *Verification:* Look for Auto-assign public IPv4 address and ensure it is set to Enable so instances launched there automatically get a public IP.



2) Enable VPC peering for cross-region.

Steps to Enable Cross-Region VPC Peering (AWS)

1.Verify Prerequisites: Check CIDR blocks.

Ensure that the two VPCs located in different regions do **not** have overlapping IPv4 or IPv6 CIDR blocks. Overlapping blocks will cause the peering request to fail immediately.

Verification: Check the VPC dashboard details for both VPC-1 and VPC-2 to confirm unique IP ranges (e.g., 10.0.0.0/16 and 172.16.0.0/16).

2.Create the Peering Connection: Requester Region.

1. Open the Amazon VPC console in the Region of your **Requester VPC**.
2. In the navigation pane, choose **Peering connections**, then click **Create peering connection**.
3. Provide a name tag for the connection.
4. Under **VPC ID (Requester)**, select your local VPC.
5. Under **Select another VPC to peer with**, choose **Another Region** and select the target AWS Region. Enter the **VPC ID (Accepter)** of the remote VPC.
6. Choose **Create peering connection**.

Verification: The peering connection status will show as Pending-acceptance.

3.Accept the Peering Connection: Accepter Region.

1. Switch your AWS Console to the **Accepter Region** specified in the previous step.
2. Navigate to **Peering connections** in the VPC console.
3. Select the pending peering request, click **Actions**, and choose **Accept request**.

Verification: The status of the peering connection changes to Active.

4.Update Route Tables: Both Regions.

To route traffic through the peering connection, update the route tables for subnets in both VPCs.

1. Go to **Route Tables** in your VPC console for Region A.
2. Select the route table associated with your subnet and choose **Edit routes** -> **Add route**.
3. For **Destination**, enter the CIDR block of the peer VPC (e.g., 172.16.0.0/16).
4. For **Target**, select **Peering connection** and choose your newly created inter-region peering connection ID. Save changes.
5. Repeat this process in Region B, pointing back to Region A's CIDR block.

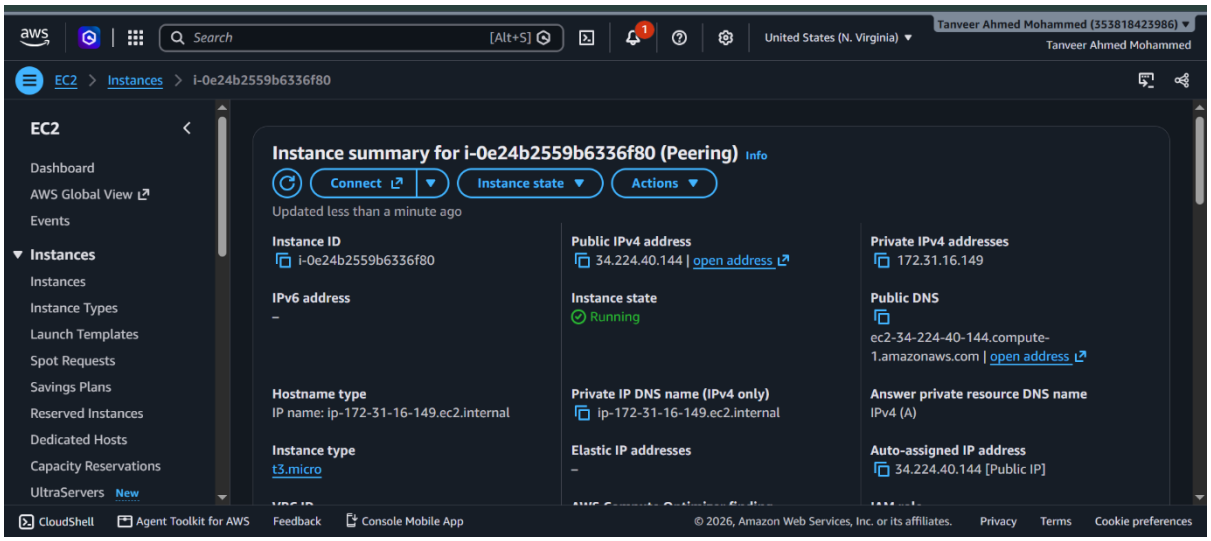
Verification: Check route table entries to confirm the peer VPC CIDR maps directly to the active peering connection target.

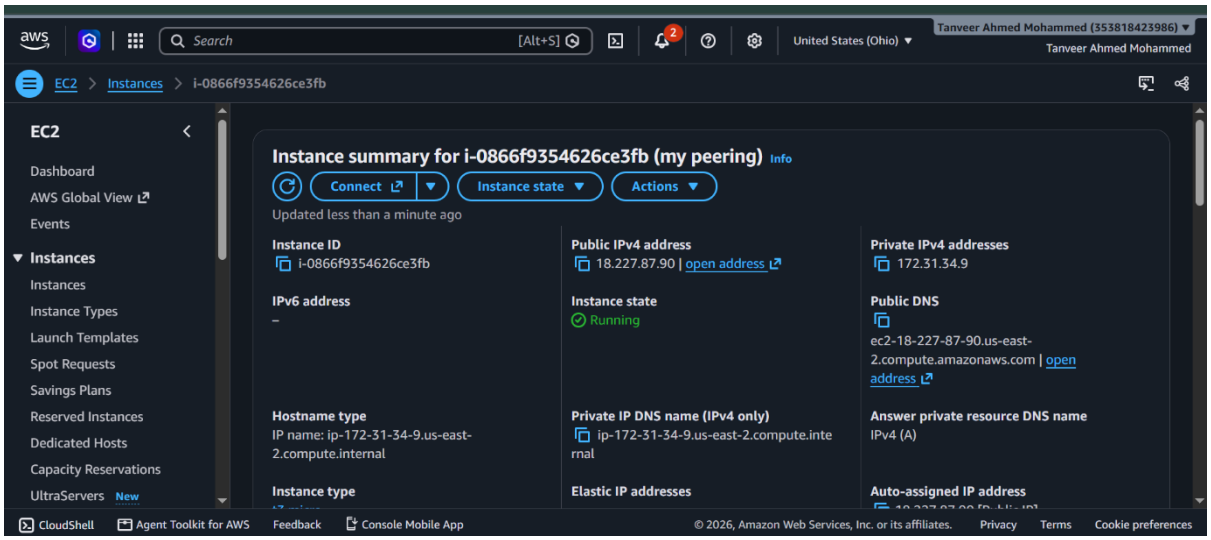
5.Configure Security Groups & Firewalls: Instance Access.

Update the security groups attached to your instances or databases in both VPCs to allow inbound traffic from the peer VPC's IP range or security group.

Verification: Run a connectivity test or ping/SSH from an instance in Region A to an instance in Region B using their private IP addresses.

Here I have Taken USA Virginia and Ohio region Both VPC is connected through same ID but Different Regions





3)Enable VPC peering for cross-account (you can collaborate with your friend to do this task).

Step-by-Step Cross-Account VPC Peering Setup

1.Verify CIDR Block Uniqueness: Prerequisite Check.

Ensure that **VPC A** and **VPC B** have completely non-overlapping IPv4 CIDR blocks (e.g., VPC A uses 10.0.0.0/16 and VPC B uses 172.31.0.0/16). If they overlap, the connection will fail immediately.

Verification: Inspect the VPC dashboard details in both AWS accounts to verify unique ranges.

2.Create an IAM Role for Acceptance: Account B (Friend's Account).

Your friend needs to create an IAM role in their account (Account B) that grants your account (Account A) permission to accept the peering request.

1. Go to the **IAM Console** in Account B and create a new role with a **Custom Trust Policy** that allows Account A's account ID to assume it.
2. Attach a permissions policy containing ec2:AcceptVpcPeeringConnection.
3. Share the resulting **Role ARN** with you.

Verification: Confirm the Role ARN is active and correctly references your requester account ID in its trust relationship.

3.Initiate the Peering Connection: Account A (Your Account).

1. Open the **Amazon VPC Console** in Account A.
2. In the navigation pane, choose **Peering connections** -> **Create peering connection**.
3. Select your local VPC as the **Requester VPC**.
4. Under **Select another VPC to peer with**, choose **Another account** and enter **Account B's Account ID**.
5. Enter **VPC B's ID** and the target Region.
6. Click **Create peering connection**.

Verification: The peering state will show as Pending acceptance.

4.Accept the Peering Request: Account B (Friend's Account).

Your friend must accept the request using the IAM role or direct console access:

1. Log into the AWS Console in Account B and navigate to **Peering connections**.
2. Select the pending request, click **Actions**, and choose **Accept request**.

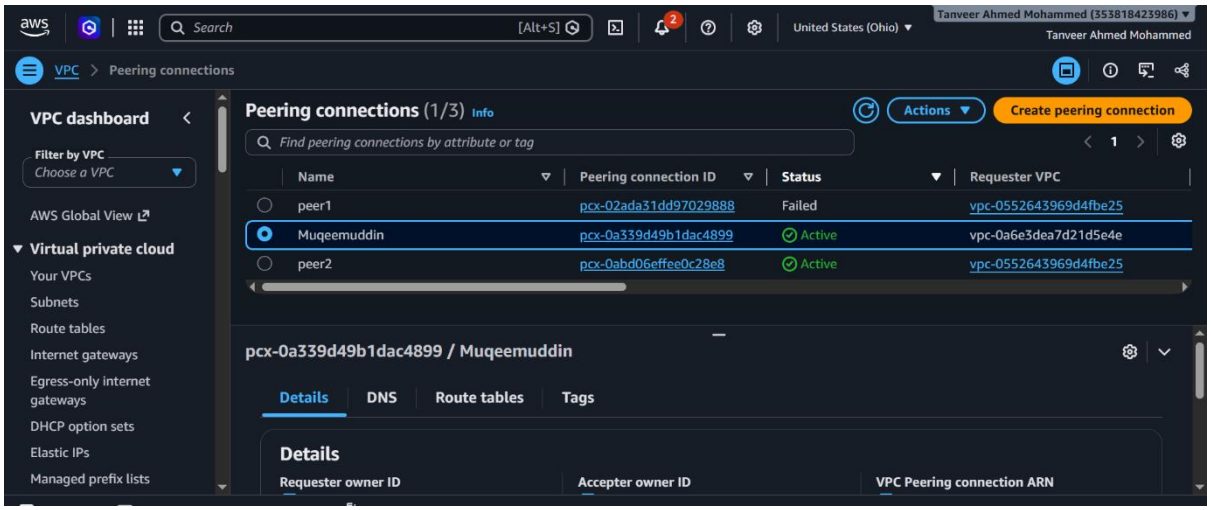
Verification: The status of the peering connection updates from Pending-acceptance to Active.

5.Update Route Tables and Security Groups: Both Accounts.

1. **Route Tables:** In both accounts, open the route tables associated with your subnets and add a route pointing the peer VPC's CIDR block to your peering connection ID (pcx-xxxxxx).

2. **Security Groups:** Update the security group rules on your instances to allow inbound traffic (such as SSH, ICMP, or custom ports) from the peer VPC's private IP range.

Verification: Run a private IP ping or connection test from an instance in VPC A to an instance in VPC B.



4) Set up a VPC Transit Gateway.

Step-by-Step Transit Gateway Setup

1.Create the Transit Gateway: Region selection.

1. Open the **Amazon VPC Console** in your chosen region.
2. In the left navigation pane, scroll down to **Transit Gateway** and click **Transit Gateways**.
3. Click **Create transit gateway**.
4. Enter a **Name** (e.g., Main-Transit-Gateway).
5. Configure settings: leave **Amazon side ASN** as default (or set a custom one if required), and check or uncheck **Auto accept shared attachments** depending on your cross-account strategy.
6. Click **Create transit gateway**.

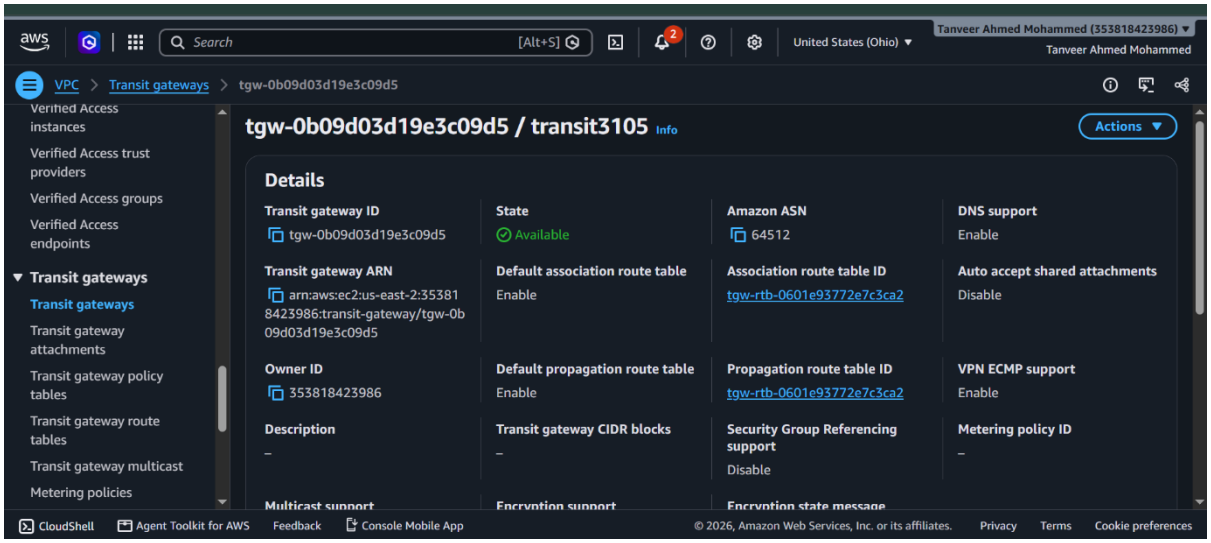
Verification: Wait until the state of the Transit Gateway changes from pending to available.

2.Create Transit Gateway Attachments: Connect VPCs.

You need to create an attachment for each VPC you want to connect to the hub.

1. In the VPC console navigation pane, click **Transit Gateway Attachments -> Create transit gateway attachment**.
2. Select your newly created **Transit Gateway ID**.
3. Under **Attachment type**, select **VPC**.
4. Select the **VPC ID** you want to attach and choose the specific **Subnets** (ideally one per Availability Zone for high availability).
5. Click **Create transit gateway attachment**.
6. Repeat this process for any other VPCs you wish to connect.

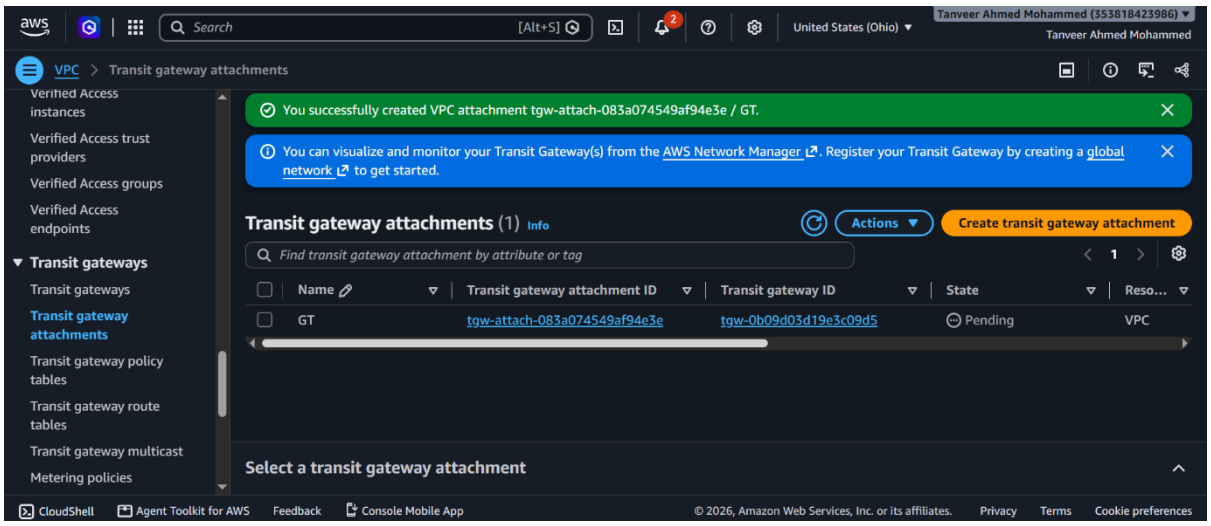
Verification: The attachment state will change from pending to available (this may take a few minutes).

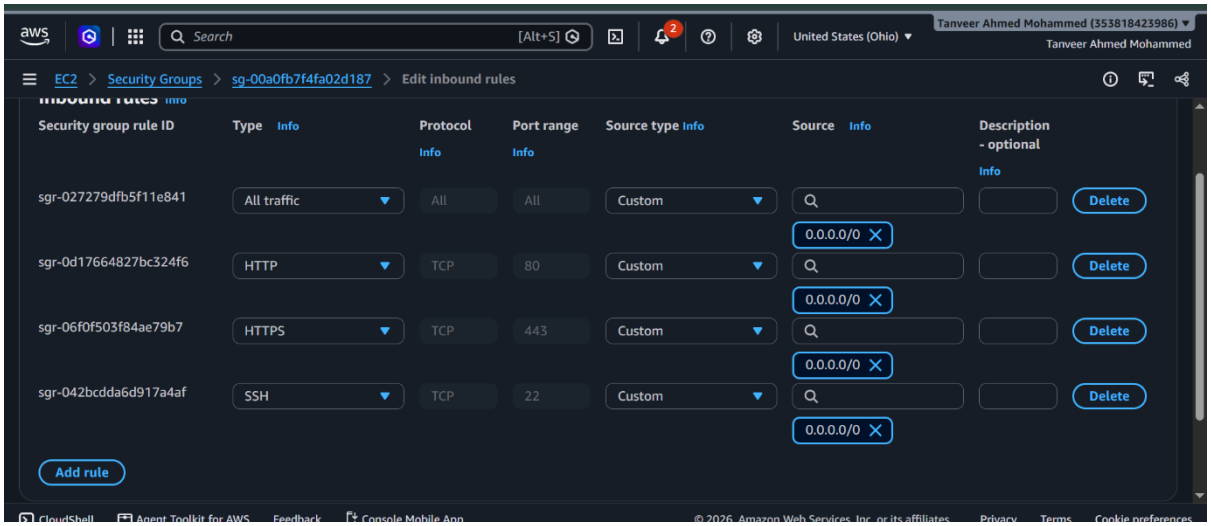


3.Configure Transit Gateway Route Tables: Route propagation or static routes.

1. In the VPC console navigation pane, click **Transit Gateway Route Tables**.
2. Select your Transit Gateway Route Table (or create a custom one).
3. Under the **Associations** tab, verify that your VPC attachments are associated.
4. Under the **Propagations** tab, enable route propagation for your attached VPCs so the Transit Gateway automatically learns their CIDR blocks.

Verification: Check the **Routes** tab within your Transit Gateway Route Table to confirm that the destination CIDRs of your attached VPCs appear automatically.





5.Configure Security Groups and Firewalls: Security group check.

Update the security groups on your EC2 instances or resources in both VPCs to allow inbound traffic from the IP ranges of the other connected VPCs.

Verification: Run a private IP ping or SSH connection test between an instance in VPC 1 and an instance in VPC 2.

5) Set up a VPC Endpoint

1.Start Endpoint Creation: Endpoints Dashboard.

1. Click the orange **Create endpoint** button in the top right corner of your screen. *Verification:* The configuration form for a new VPC endpoint should open.

2.Configure Service Details: Service Selection.

1. Enter a **Name tag** (e.g., ssm-interface-endpoint).
2. Under **Service category**, select **AWS services**.
3. In the **Service** search box, select the specific service you want to connect to (make sure it ends with .interface, such as com.amazonaws.us-east-1.ssm).

Verification: The selected service displays correctly under the service list.

3.Select VPC and Subnets: VPC and Subnets.

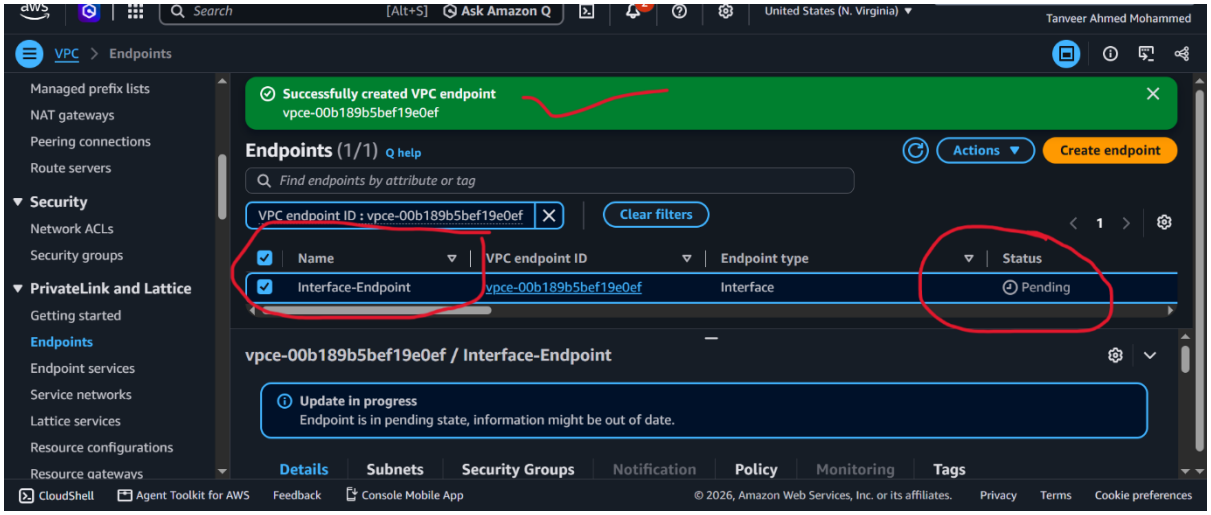
1. Under **VPC**, choose your target custom VPC or default VPC.
2. Select the **Subnets** where your EC2 instances reside so AWS can deploy the endpoint network interfaces (ENIs) there.

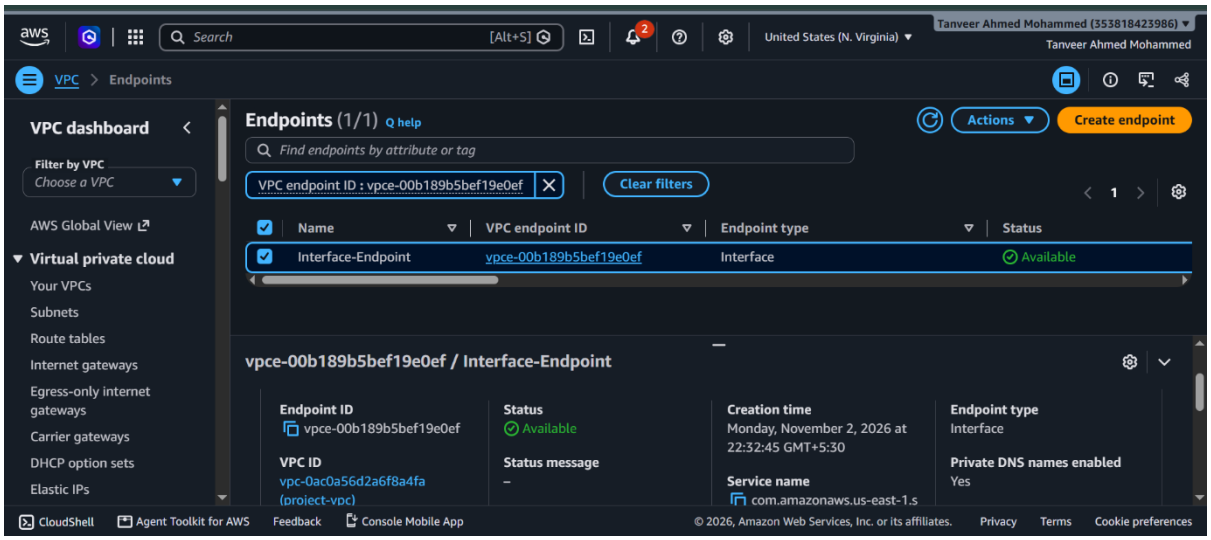
Verification: Selected subnets match the availability zones of your running instances.

4.Attach Security Groups and Create: Security Groups & Completion.

1. Under **Security groups**, select a security group that allows inbound HTTPS (Port 443) traffic from your instances.
2. Check **Enable private DNS name** (if applicable) and click **Create endpoint** at the bottom right.

Verification: The endpoint status changes from Pending to Available after a couple of minutes.





Here, Its Interface Endpoint is Active!